Recap 1 · TinyML: machine learning on the sensor

Three applications we built

AUDIO Day 3

Keyword spotting

"Hey Nicla": wake-word detection

VISION Day 1

Object classification

What is in the image?

VISION Day 1

Object detection

What, where and how many? (FOMO)

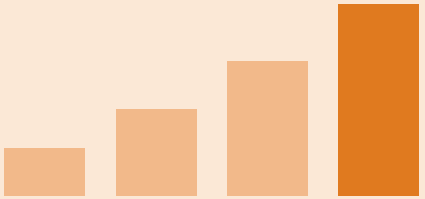
The workflow, every time



Tools **Edge Impulse** collect, label, train **Arduino IDE** build and flash

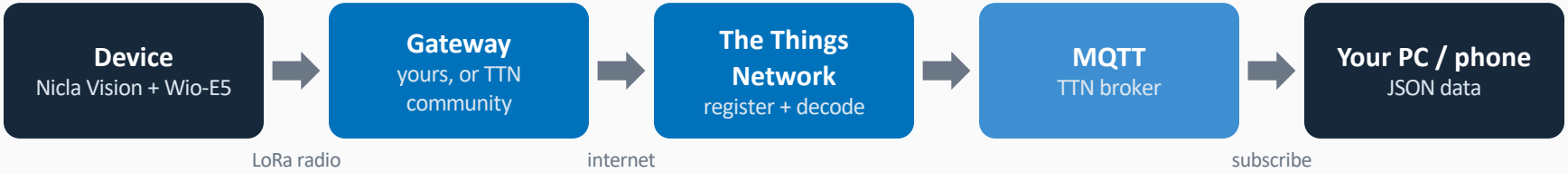
Main challenge

You need a lot of data!



Many labelled samples for every class, balanced and recorded in real conditions.

Recap 2 · LoRaWAN: from the sensor to your screen



1 · Gateway

Install a new gateway…

…or use **TTN** community gateways **for free**.

2 · Register on TTN

A new device needs three keys:

- **DevEUI**
- **JoinEUI (AppEUI)**
- **AppKey**

Copy them into your sketch.

3 · Payload decoder

TTN needs a decoder written in **JavaScript**

```
function decodeUplink(p)
{
  var b = p.bytes;
  // bytes → values
  return { data: {…} };
}
```

bytes in → readable fields out

4 · Get the data with MQTT

The topic is long:

```
v3/<app-id>@ttn/devices/<dev-id>/up
```

Tip: subscribe from your PC first (topic #) to find it.

The final packet is JSON:

```
{ "end_device_ids": {…},
  "uplink_message": {
    "decoded_payload": {
      "value": 42 } } }
```